

A Local-Obstruction Counterexample to Conjecture 000000000008

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Statement of the conjecture

Conjecture 000000000008 concerns the number of prime values of a square-free integer-coefficient quadratic polynomial. It asserts that there is a constant $c_f > 0$ such that the prime-value count has an explicit lower bound of order

$$c_f \frac{x}{\log x}.$$

The statement, as written, does not impose the usual local-obstruction condition that no prime divide every value of f .

Counterexample

Consider

$$f(n) = n^2 + n + 2 = n(n + 1) + 2.$$

This is an integer-coefficient quadratic polynomial. Its discriminant is

$$\Delta = 1^2 - 4 \cdot 1 \cdot 2 = -7 \neq 0,$$

so it has no repeated root.

For every integer n , one of n and $n + 1$ is even. Hence $n(n + 1)$ is even, and therefore $f(n)$ is even. For every positive integer n ,

$$f(n) = n^2 + n + 2 \geq 4.$$

Consequently, for every positive integer n , $f(n)$ is an even integer strictly larger than 2, and therefore is not prime.

Thus the polynomial has no prime values on the positive integers. If $\Pi_f(x)$ denotes the number of positive integers $n \leq x$ for which $f(n)$ is prime, then

$$\Pi_f(x) = 0$$

for every positive integer x . This contradicts any lower bound

$$\Pi_f(x) \geq c_f \frac{x}{\log x}$$

with $c_f > 0$ for sufficiently large x .

What the counterexample shows

The failure is caused by a fixed local obstruction at the prime 2:

$$f(n) \equiv 0 \pmod{2} \quad \text{for every integer } n.$$

A corrected version of the conjecture would need, at minimum, to exclude polynomials having a fixed prime divisor. This condition is separate from having nonzero discriminant: the present polynomial has discriminant -7 , but still has the fixed prime divisor 2 in all its values.

Formal verification

The accompanying Lean 4 project defines the polynomial, the discriminant, a fixed-divisor-free elementary prime predicate, and the prime-value count on positive inputs. It proves the following facts:

- $f(n) \bmod 2 = 0$ for every natural number n ;
- $f(n) \geq 4$ for every positive natural number n ;
- no positive input gives a prime value;
- the discriminant is nonzero;
- the prime-value count is identically zero.

The project uses Lean 4.33.1 and only the standard library. It contains no axioms or admitted proof steps.